

Maximum Nullity of Cyclic Covers of Graphs

A monodromy ceiling, and the zero forcing number of $P(n, k)$ determined for *every* large n when $k = 2, 3, 4$

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Every claim below is labelled proved / certified / numerical. All code, certificates and the dated research log are open.

1. The problem

For a graph G , colour a set S of vertices. If a coloured vertex has exactly one uncoloured neighbour, that neighbour becomes coloured; repeat. $Z(G)$ is the least $|S|$ that eventually colours everything. Z bounds the *maximum nullity* $M(G)$, the largest null space among symmetric matrices whose off-diagonal support is exactly $E(G)$:

$$M(G) \leq Z(G).$$

The *generalized Petersen graph* $P(n, k)$ has outer cycle $u_i u_{i+1}$, inner edges $v_i v_{i+k}$ and spokes $u_i v_i$; cubic on $2n$ vertices, with the rotation $\rho: i \mapsto i + 1$. More generally $P(n, k)$ is a *cyclic cover*: a base graph with edge voltages in $\mathbb{Z}_n$, lifted to $V(\mathcal{B}) \times \mathbb{Z}_n$. Two vertices, loops of voltage 1 and k , a spoke of voltage 0. Everything below is really about covers.

2. What was open

Rashidi, Shajareh Poursalavati and Tavakkoli [1] proved $Z(P(n, k)) \leq 2k + 2$ and $Z(P(n, 2)) = 6$ for $n \geq 10$. Their Theorem 3.6, claiming $Z(P(n, 3)) = 8$ for $n \geq 12$, is *false*: Krishnan [2] exhibits $Z(P(12, 3)) = 7$, the flaw being a case analysis that never excludes 7-vertex sets. That July 2026 note closes by recording as open

the stabilization threshold, and a rigorous lower bound for $k \geq 4$.

Lower bounds are the hard direction: an upper bound exhibits one forcing set, a lower bound must exclude *every* smaller one.

3. Why the standard tools fail

A set is forcing iff it meets every *fort*. All three standard consequences degrade as k grows, and are insufficient *in principle*:

- ▶ disjoint forts: at most *two* exist in any case tested;
- ▶ fort LP: integrality gap ≈ 2 (LP = 3.21 against $Z = 10$ for $P(18, 4)$), which also explains why fort solvers scale poorly here;
- ▶ spectral: Hoffman-type bounds give 1.2–2.9 against true Z of 6–12.

This is what motivated changing the *category* of argument, from combinatorial to matrix-analytic.

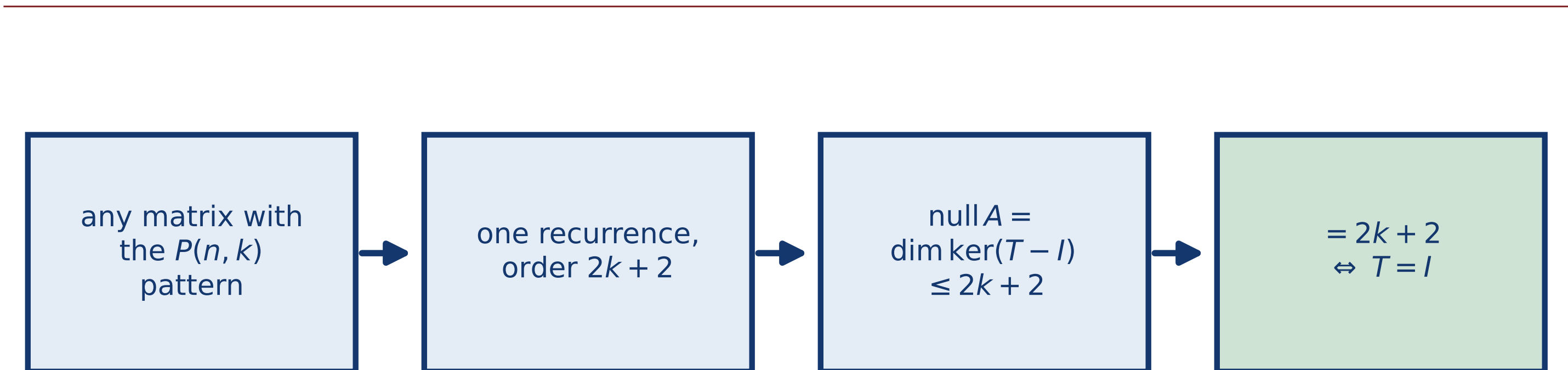
4. The ceiling

Every spoke weight c_i is nonzero, so the kernel equation at u_i solves for $y_i = -(b'_{i-1}x_{i-1} + a_i x_i + b_i x_{i+1})/c_i$. Substituting into the equation at v_i eliminates y entirely and leaves *one scalar recurrence* in x , supported on nine indices, with both end coefficients $\gamma_{i,\pm(k+1)} \neq 0$.

Theorem 1. *Let $n \geq 2k + 3$ and let A have off-diagonal support exactly $E(P(n, k))$ — symmetric or not. Then $\text{null } A = \dim \ker(T - I) \leq 2k + 2$, where T is the monodromy of that order- $(2k + 2)$ recurrence, with equality iff $T = I$.*

The hypothesis is sharp: at $n = 2k + 1$, $2k + 2$ the offsets $\pm(k + 1)$ collide modulo n and the order drops.

The order $2k + 2$ is exactly the size of the forcing set the rotation bootstrap produces. **The ceiling of the matrix method equals the upper bound it is trying to match** — so either the method settles Z completely, or nothing does.



the ceiling equals the upper bound: no slack either way

Figure 1. The reduction, and where the ceiling comes from.

5. Reaching the ceiling: the symbol

For A invariant under ρ the Fourier blocks are $M_m = \begin{pmatrix} z^{+5m} & z^{-c} \\ c & d + e z_m \end{pmatrix}$, $s_m = 2 \cos \frac{2\pi m}{n}$. Since $t = L_k(s)$ exactly, with L_k the monic integer polynomial satisfying $L_k(z + z^{-1}) = z^k + z^{-k}$,

$$\text{null } A = \#\{m : F(s_m) = 0\}, \quad F(s) = (s + a)L_k(s) + \alpha s + \beta,$$

monic of degree $k + 1$ but carrying *exactly three* free parameters, for every k . Three roots may be prescribed and no more — but prescribing three *determines the other* $k - 2$, and if F is rational its roots form Galois orbits, which stay inside the grid $\Gamma_n = \{2 \cos(2\pi m/n)\}$. One rational F places all $k + 1$ roots at once. Taking $F = \prod_{d \in D} \Psi_d$, a product of minimal polynomials of $2 \cos(2\pi/d)$, gives *integer* certificates.

Theorem 2. *Realisability is a factorisation: $c^2 = a\alpha - \beta = 0$ if and only if $F = (s + a)(L_k(s) + t)$. Over $\mathbb{Q}$, Niven’s theorem forces $a, t \in \{0, \pm 1, \pm 2\}$, so at most 25 degenerate symbols exist for each k .*

Only $c^2 = 0$ is fatal: c^2 is the *product* of the two spoke weights, so $c^2 < 0$ is realisable once they may differ. Imposing $c^2 > 0$ instead — our own earlier error — wrongly ruled out $k = 3$ and $k = 7$.

6. Why congruence conditions are forced

Every certificate above holds only for n in a divisibility class. That is not a shortfall in the search:

Theorem 3. *A period-one symbol is a fixed polynomial; its roots are fixed algebraic numbers, and $2 \cos(2\pi j/d) \in \Gamma_n$ iff $d \mid n$. So one such matrix settles exactly $\{n : \text{lcm}(D) \mid n\}$ — a single class, containing **at most one prime**. For $k \geq 3$ every symbol obeys $e_2 = -k$, a Conway–Jones relation on the grid.*

For $k = 2$ that coefficient is free, *which is exactly why $k = 2$ alone was ever general in n* . Escaping the classes therefore requires abandoning rotation-invariance altogether — and Theorem 1 characterises the ceiling for *every* matrix on the pattern by $T = I$, a condition with no arithmetic in it.

7. Tiling: all large n

Build the weight sequence out of pieces. Pieces do not compose naively, since T_i depends on weights within $2k + 2$ of position i ; so pin a *docking pattern* on the first and last $k + 1$ positions of every tile.

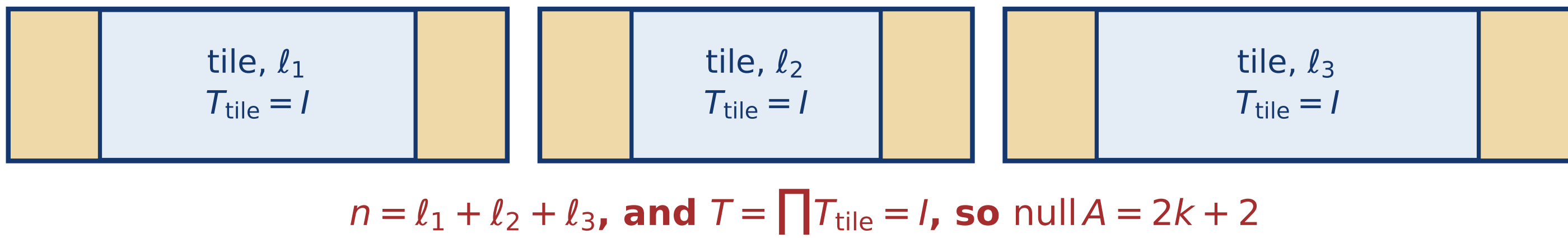
Lemma 4. *Each tile’s transfer product then depends only on its own interior, and $T = \Pi T_{\text{tile}}$.*

Theorem 5. *If identity tiles exist for every length in $[L, 2L]$, then $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for every $n \geq L$.*

Proof. Every $n \geq L$ is a sum of lengths drawn from $[L, 2L]$: take one tile if $n < 2L$, else remove a tile of length L and recurse. The monodromy is the product of the tile products, hence I ; apply Theorem 1. $\square$

A full interval is what makes this gapless. Two coprime lengths a, b would give only $n = ia + jb$, leaving every $n < (a - 1)(b - 1)$ uncovered — several hundred at the smallest solvable lengths.

tiles of every length in $[L, 2L] \Rightarrow$ every $n \geq L$ is a sum of them



shaded ends: the fixed docking pattern of $k + 1$ positions, which makes each tile’s product depend on its interior alone

Figure 2. Tiles compose because their ends are pinned; all large n becomes a finite list of finite problems.

8. Certifying a tile

$T_{\text{tile}} = I$ is *rational* in the weights, so a Krawczyk bound on it would need interval matrix products. It is unnecessary: by Theorem 1 the monodromy of the *one-tile* cyclic matrix *is* the tile product, so

$$T_{\text{tile}} = I \iff \text{null } A_{\text{one-tile}} = 2k + 2,$$

and a nullity is certified by the **bilinear** system $A(w)K(X) = 0$, whose Jacobian is affine and whose Lipschitz constant is an explicit integer.

$AK = 0$ has rank $2nr - \binom{r}{2}$, and the deficiency is structural: $K'AK$ is symmetric because A is, and $\binom{r}{2}$ is its antisymmetric part. Choosing the square subsystem *structurally* — non-pivot rows plus the upper triangle of the pivot block — makes it *equivalent* to $AK = 0$: choosing it by pivoting leaves $\binom{r}{2}$ equations unproved and the Krawczyk test still passes. Those equations fall to 10^{-61} without being solved for.

9. Results

Theorem 6. *With no congruence condition:*

$$\begin{aligned} Z(P(n, 2)) &= M(P(n, 2)) = 6 && \text{for all } n \geq 9, n \neq 10, \\ Z(P(n, 3)) &= M(P(n, 3)) = 8 && \text{for all } n \geq 17, \\ Z(P(n, 4)) &= M(P(n, 4)) = 10 && \text{for all } n \geq 30. \end{aligned}$$

Theorem 7. $Z(P(n, 2))$ and $Z(P(n, 3))$ are determined for every admissible n . In particular $Z(P(n, 3)) = 8$ for every $n \geq 13$, and 13 is optimal: $Z(P(11, 3)) = Z(P(12, 3)) = 7$.

This determines the **stabilization threshold** for $k = 3$ — one of the two problems [2] records as open — and the exact values for $k = 4$ answer the other. The sequence is genuinely non-monotone below the threshold: $Z(P(n, 3)) = 6, 6, 6, 8, 7, 7, 8, 8, \dots$ for $n = 7, \dots, 14$.

Which n are settled: sparse congruence classes (blue) against every n past a threshold (green)

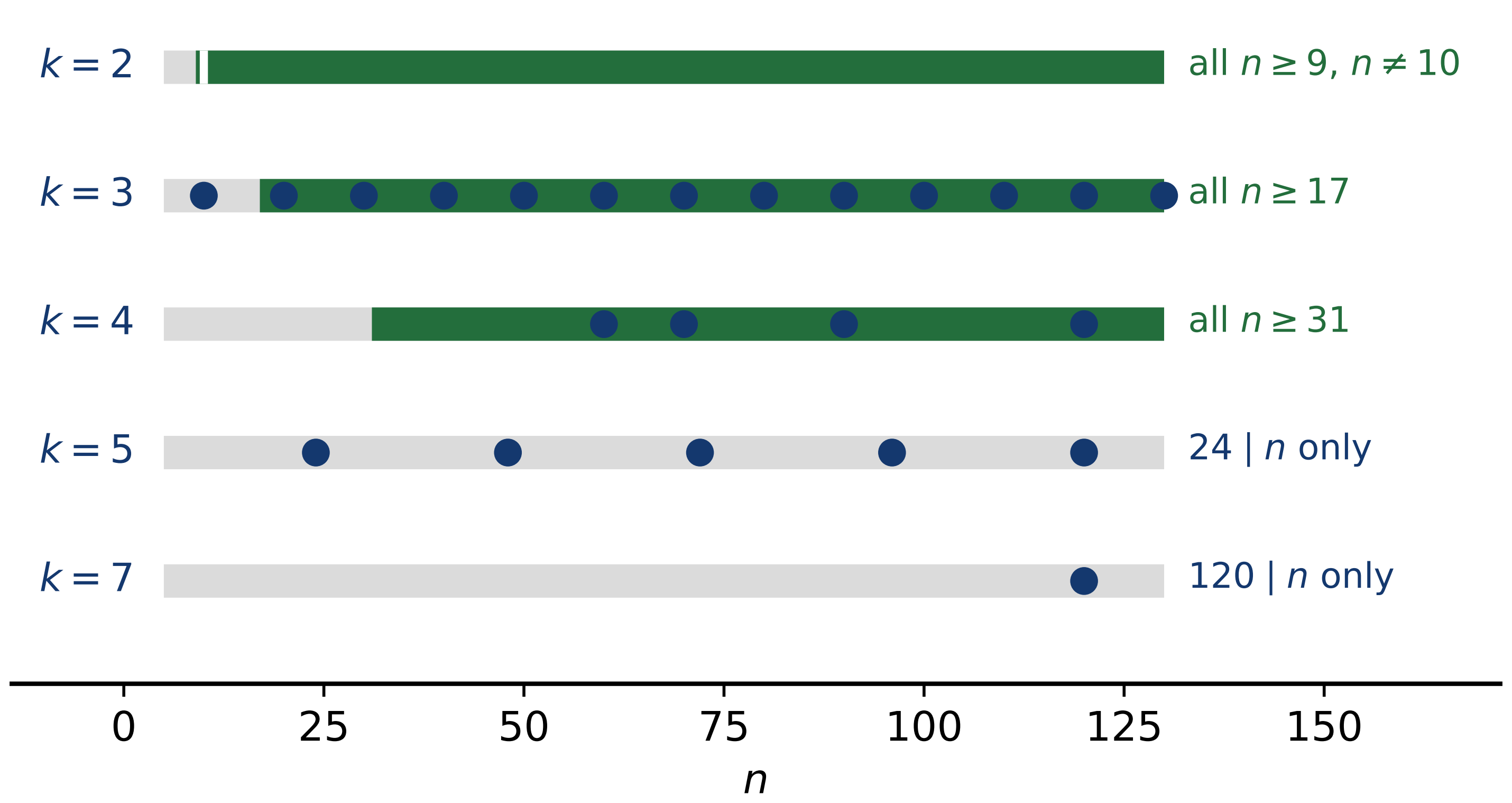


Figure 3. Sparse congruence classes (dots) against every n past a threshold (bars).

10. How each is proved

$k = 2$: a closed form. Matching coefficients gives $c^2 = -(s_1 + s_2)(s_1 + s_3)(s_2 + s_3)$, so $c^2 = 0$ iff two roots are antipodal; the three most negative grid values make it positive for all $n \geq 9$, $n \neq 10$.

$k = 3, 4$: the tiling theorem with $L = 21$ and $L = 31$, from **54 certified identity tiles** covering $[21, 43]$ and $[31, 61]$. Below L , $k = 3$ is closed by 17 consecutive individually certified values, $17 \leq n \leq 33$.

k	settles	L	tiles	max α
2	$n \geq 9, n \neq 10$	—	closed form	—
3	$n \geq 17$	21	23	2.1×10^{-10}
4	$n \geq 30$	30	32	3.8×10^{-10}

Each tile is an independent Krawczyk contraction with its own unknown count, $\|Y\|_\infty$ and residual; nothing is copied between them. At $k = 4$, $\ell = 61$: **1425** unknowns, **1220** equations of which **1175** are structural and **45** = $\binom{10}{2}$ dependent, high-precision residual 4.7×10^{-61} .

11. Verification

One script re-derives everything: **verify_all.py, 69 checks, all passing** — the realisability theorem symbolically for $k = 2..10$, every integer certificate, all **54** tiles re-evaluated for nullity *and* exact docking, and the claim that every n in range is a sum of certified lengths. The structural theorem was checked on **189** random matrices over **63** pairs (n, k) . The exhaustive Z solver reproduced every published value first, including the corrected $Z(P(12, 3)) = 7$.

The $n = 17$ certificate was found three independent ways — minimising $\|T - I\|$, an alternating solve, a gauge-fixed search — all yielding a certified matrix. Stage 1 is only a search; stages 2–3 are the proof.

Why a failed search proves nothing here. Every symmetric matrix on the pattern is gauge-equivalent to one in the slice we search, so a failure inside it measures effort, not impossibility. $P(18, 3)$ and the tile $\ell = 21$ were both reported as failures and both succeeded once the restart budget was raised. *Working rule: quote the residual and the budget, never just “failed”.*

12. Limits, and what is not proved

$k \geq 5$ is *not* settled: tiles exist at $k = 5$ from $\ell \approx 54$, but **[54, 108]** needs **54** of them and is incomplete. $7 \leq n \leq 12$ at $k = 3$ and $9 \leq n \leq 30$ at $k = 4$ are open. The $k \geq 6$ congruence-class values remain numerical. The rank law is verified over a range, not proved.

We first searched $k = 5$ over $41 \leq \ell \leq 45$, the parameter count’s bound, found nothing, and later found tiles starting near **54**: the count is a lower bound, *not* a predictor of where tiles begin, and we cannot explain the gap. Among claims we have withdrawn: an unconditional cover bound ($Z(P(8, 2)) = 5$ refutes it); the threshold $(2k + 2)^2/5$; $N(k) = O(k)$; and an eigenvalue search whose negative results we retracted after it reported nullity 7 at $P(20, 3)$ where 8 is attained.

References

- [1] Rashidi, Shajareh Poursalavati, Tavakkoli, *Computing the zero forcing number for generalized Petersen graphs*, J. Algebra Combin. Discrete Struct. Appl. 7(2) (2020) 183–193.
- [2] Krishnan, *A correction to the zero forcing number of the generalized Petersen graphs $P(n, 3)$* , arXiv:2607.19412 (2026).
- [3] AIM Minimum Rank Special Graphs Work Group, *Zero forcing sets and the minimum rank of graphs*, Linear Algebra Appl. **428** (2008) 1628–1648.
- [4] Conway, Jones, *Trigonometric diophantine equations*, Acta Arith. **30** (1976) 229–240.